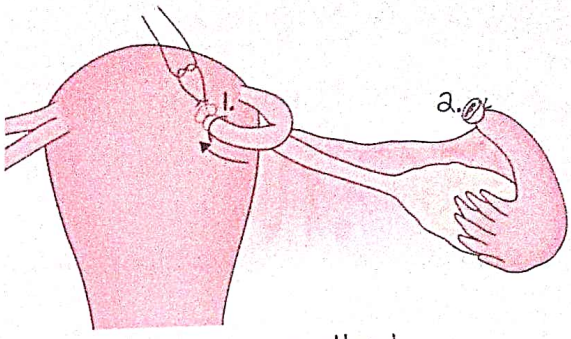
----- Active space -----

Irving method :

1. Proximal part is buried in myometrium.
2. Distal part is ligated.

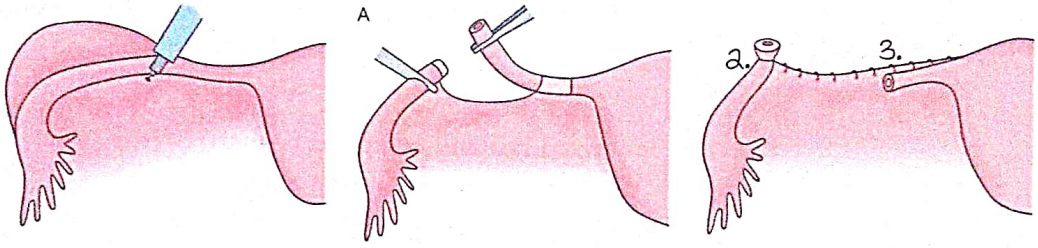


Irving method

Uchida's technique :

1. Hydrodissection : Saline injected into mesentery of fallopian tube → clear demarcation of muscularis layers.
2. Proximal end is ligated.
3. Distal end is ligated and sutured along mesentery.

I.



Uchida's technique

Madlener technique :

- Base of the tube is ligated and tube is crushed.
- Failure rate : High.

Kroener surgery :

Fimbriectomy → Failure of oocyte pick-up.

FAILURE RATE

Least ← Cautery, Uchida, Irving, Modified Pomeroy, Pomeroy → most

CHANCES OF RECANALIZATION

most ← Clips, Fallope rings, Pomeroy, Modified Pomeroy, Irving, Uchida, Cautery → Least

ENDOMETRIAL HYPERPLASIA

----- Active space -----

Overview of endometrial hyperplasia (EH):

- Histopathology diagnosis.
- Cause: ↑estrogen (Hyperestrogenic state).
- Presentation: Perimenopausal female c/o excessive bleeding.
- Classification: AUB – M.
- Investigations:
  - 1st Ix (After r/o pregnancy): TVS → ↑ ET → Suspect EH → Take → EB.
  - IOC for endometrial pathology: Endometrial biopsy (EB).
  - Gold standard: Fractional curettage + Hysteroscopy.

ET: Endometrial thickness.
TVS: Trans-vaginal sonography.

Investigations for EH

00:04:35

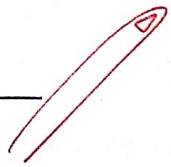
ENDOMETRIAL SAMPLING

AKA Endometrial biopsy/Endometrial aspiration cytology.

Done in: OPD.

Anaesthesia: Not required.

Instrument used →
- India: Karman's cannula
- Pipelle.
- Vabra aspirator.



Indications:

Reproductive age (<40yrs):
- C/o AUB + Risk factors of Ca Endometrium: PCOS, DM, HTN, Tamoxifen.
- ET ≥ 12mm.
- AUB unresponsive to medical Mx (Tranexamic acid, OCP, progesterone).

Perimenopausal (≥40 yrs: FOGSI/≥45 yrs: ACOG): EB irrespective of USG findings.

Menopausal:
- ET ≥ 4mm on TVS.
- Pap smear: AGCUS (Atypical glandular cells of undetermined significance).

----- Active space -----

FRACTIONAL CURETTAGE (FC)

Done in : OT.

Anesthesia : ↓GA (under general anaesthesia).

Procedure : Endocervical curettage (1st).

↓

Endometrial curettage.

Advantage : Diagnosis generalised endometrial pathology.

Disadvantage : Not good for localised pathology.

Note : Single best Ix → Fractional curettage.

HYSTEROSCOPY

used for localised pathology.

EB Report Interpretation & Management

00:14:10

EB outcomes :

1. Normal
2. metropathia hemorrhagica
3. Endometrial hyperplasia ± Atypia

1. NORMAL

- use 1st line drugs to mx AUB → Tranexamic acid
  - → OCP.
  - → Progesterone → Oral.
    - → mirena.

unresponsive ↓

FC + Hysteroscopy.

2. METROPATHIA HEMORRHAGICA

m/c : Perimenopausal female.

Pathology : Anovulatory cycles.

Amenorrhea —(x 2-3 months)→ HMB (Heavy menstrual bleeding).

HPE :

- Swiss cheese pattern.
- Proliferative endometrium.

Equivalent to cystic glandular hyperplasia.

Risk of malignancy : 1%

mx :

TOC : Progesterone → Oral
- → mirena

3. ENDOMETRIAL HYPERPLASIA

----- Active space -----

Without atypia :

- Risk of malignancy : 1–3%

↓ Next step

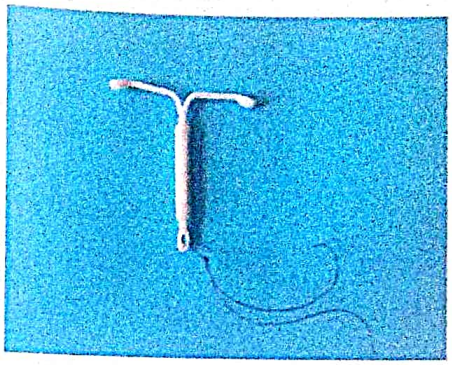
Progesterone —Downregulation of estrogen receptors→ Endometrial atrophy.

- mirena > Oral : medroxy progesterone acetate

↓

Continuous > Cyclical

- ≥ 12 days for a protective effect.
- No contraceptive benefit.



IUCD mirena

- Repeat biopsy at 3–6 months.

↓ Hyperplasia (+)/Bleeding (+)

FC + Hysteroscopy.

With atypia :

- Risk of malignancy : 8–30% (↑↑Chance).

↓ Next step

FC + hysteroscopy.

Management :

- Best : TAH (Total abdominal hysterectomy → Uterus + cervix).
- Patient refuses TAH : Progesterone (megestrol).

↓ Failure/continued bleeding

Hysterectomy.

Note :

Indications for FC + Hysteroscopy :

- EB : EH + Atypia.
- EB report : Insufficient sample.
- Incase of cervical stenosis.
- EB : EH w/o atypia + failure of medical mx.

----- Active space -----

ENDOMETRIAL CANCER

00:01:03

Associated Factors & Screening

Seen in hyperestrogenic states.

ASSOCIATED FACTORS

Risk factors :

Mnemonic : Family has old aunt.

- Familial syndromes.
- Hypertension.
- Obesity : m/c.
- Late menopause & early menarche.
- Diabetes.
- Atypical endometrial hyperplasia.
- Unopposed estrogen.
  - PCOS.
  - Estrogen only HRT.
  - Estrogen secreting hormones. (Granulosa cell tumours).
- Nulliparity.
- Tamoxifen : and m/c.
  - Time between tamoxifen & pregnancy → Min : 2 months. / Ideal : 3 months.

Protective factors :

- multiparity.
- Pregnancy.
- OCP : Progesterone downregulates estrogen receptors on endometrium → ↓Endometrial proliferation → Endometrial atrophy.

Proposed protective factors : Physical exercise and green tea.

Corpus cancer syndrome :

Hypertension.
Obesity.
Diabetes.
} ↑risk of endometrial Ca.

Familial syndromes :

1) Lynch II syndrome :
   - A.K.a. HNPCC syndrome : Hereditary non polyposis colorectal carcinoma.
   - Cancers associated with Lynch II :
     - m/c : Colon cancer.

- and m/c : Endometrial Ca > Ovarian Ca.
- Genes involved : MLH_1, MSH_2, MSH_6.
- 60% chance of endometrial ca. in Lynch syndrome.

2) Cowden syndrome : mutation in PTEN gene on chromosome 10.

3) BRCA 1 & 2 mutation :
- Associated cancers : Ovarian ca > Endometrial ca.

----- Active space -----

SCREENING

Universal screening not done.

Indication : Hereditary cancer.

Annual endometrial biopsy :
- From 35 years of age.
- Mx : OCPs if needed.
- Best method of prevention in familial syndromes : TAH + BSO
  - TAH : Prevents endometrial ca.
  - BSO : Prevents ovarian ca.
  - Indication : Completed family by 40 years of age.

Features of Endometrial Cancer

00:11:44

HISTOPATHOLOGIC TYPES
- Adenocarcinoma : m/c type.
  - Uterus lined by columnar epithelium.
  - A.K.a. endometrioid tumour.
- Clear cell : most malignant (Clear cell > Papillary serous).
- Serous.
- mucinous.
- Carcinosarcoma.

Grading of adenocarcinoma :

Based on undifferentiated solid component.

Grade	% of solid component	Differentiation
1	<5%	Well differentiated.
2	6% – 50%	moderately differentiated.
3	≥50%	Poorly differentiated.

----- Active space -----

Types of endometrial carcinoma :

	Type 1	Type 2
Types	• (m/c) • Adenocarcinoma grade 1, 2.	• Adenocarcinoma grade 3. • Clear cell. • Papillary serous.
Prognosis	Good.	Bad.
Associated features	↑Estrogen.	↓Estrogen
Predisposing factors	Endometrial hyperplasia.	Endometrial atrophy.
Age & features	• 50 – 60 years. • Obese.	• 60 – 70 years. • Thin.
Gene mutation	• PTEN • KRAS	p53 gene

Age group of endometrial cancer
- m/c : 50 – 70 years.
- median : 60 years

Presentation :

- Irregular vaginal bleeding/AUB : m/c complaint.
- Post menopausal bleeding (PMB) : most specific.
  - 10% of PMB → Endometrial ca.
  - m/c cause of PMB : Polyp (37%) > Endometrial atrophy (30%).
- Dirty vaginal discharge.
- Pelvic pressure/pain.
- Pain referred to right hypochondrium : Simpson pain.

Spread :

Direct : m/c route.

Note :

Gatekeeper gene : PTEN.

Management

00:21:26

DIAGNOSTIC ALGORITHM

TVS : First investigation

↓

Endometrial thickness ≥4 mm

↓

Endometrial biopsy : IOC

↓ malignancy +ve

Fractional curettage + Hysteroscopy → Gold standard :
- Assesses type, grade & histology

----- Active space -----

If required :

MRI (Best investigation) : Involvement of myometrium & parametrium.

CT (Best investigation for omental mets) : Involvement of lymph nodes.

PET-CT : Best investiagtion for lymph node status.

↓

Surgical staging

↓

Histopathological examination

↓

Staging of tumour

↓

Post-op mx

mx of choice : Radiotherapy.

STAGING

FIGO 2009 :

Stage		Feature
1	Limited to uterus.	
	1A	<50% of myometrium involvement.
	1B	>50% of myometrium involvement.
2	Involvement of cervical stroma and glands.	
3		Beyond cervix.
	3A	Serosa & adnexa (Fallopian tube & ovaries) involvement.
	3B	Vagina/parametrium involvement.
	3C	Lymph node involvement.
	$3C_1$	Pelvic nodes.
	$3C_2$	Para-aortic nodes.
4	metastasis.	
	4A	Regional metastasis to bladder/bowel.
	4B	Distant metastasis or superficial lymph node involvement.

Superficial lymph nodes involvement - Stage 4B :

In endometrial ca, ovarian ca and cervical ca.

Endometrial ca :

Spread to cervix - Stage I → II.

Cancer cervix :

Spread to uterus : Stage does not change.

2023 Staging :

Surgical staging + Histopathological staging + molecular classification.

----- Active space -----

Molecular classification :

- Good prognosis : POLE mutation.
- Intermediate prognosis : mismatch repair defect (mmRD), microsatellite instability (MSI).
- Bad prognosis : p53 abnormality.

Classification based on prognosis :

	Good prognosis	Bad prognosis
Type	1	2
Grade	1, 2	3
Spread	-	To cervix or structures beyond myometrium involvement ≥50%.
mutation	POLE mutation.	p53.
ER/PR status	ER +ve, PR +ve.	ER -ve, PR -ve.
Stage	Endometroid tumour of uterus + Ovary simultaneously : Stage IA3.	Spread of uterine tumour to ovary : Stage 3A.

Surgical staging :

Surgery :

Based on stage :

- Stage 1 : TAH + BSO (Type 1 Hysterectomy).
- Stages 2 & 3 : Wertheim's Hysterectomy (Type 2 Hysterectomy).
- Stage 4 : Debulking Sx.

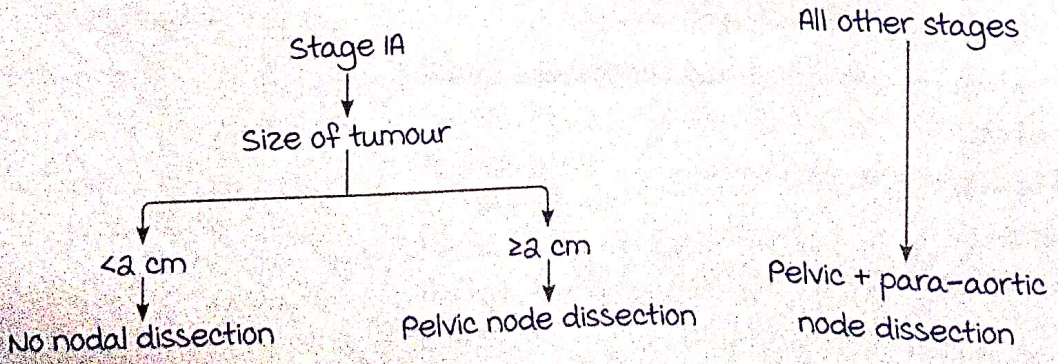
In papillary serous tumour/Carcinoid sarcoma :

TAH + BSO + Intracolic omentectomy → D/t ↑ chances of micrometastasis of omentum

In mucinous tumour : TAH + BSO + Appendicectomy.

- D/t association with appendix cancer.

Lymph node dissection :



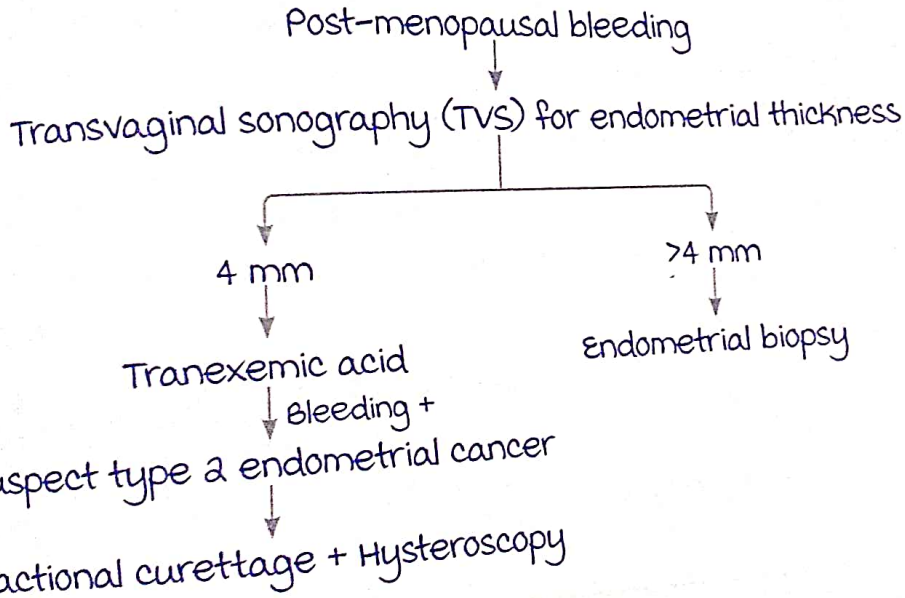
post-op management :

Risk	Includes	Post-op mx
Low	Adenocarcinoma grade 1/2 + Stage IA.	No mx.
High	Stage 3/4 Or Papillary serous/Clear cell tumour.	Chemotherapy + Radiotherapy : Carboplatin + Paclitaxel.
Intermediate	All others.	Radiotherapy.

----- Active space -----

Post-menopausal Bleeding

00:49:08



Causes :

- Polyp (m/c).
- Endometrial atrophy (and m/c).
- HRT.
- Endometrial Ca (10%).
- Cancer cervix : m/c cause of PMB in India.

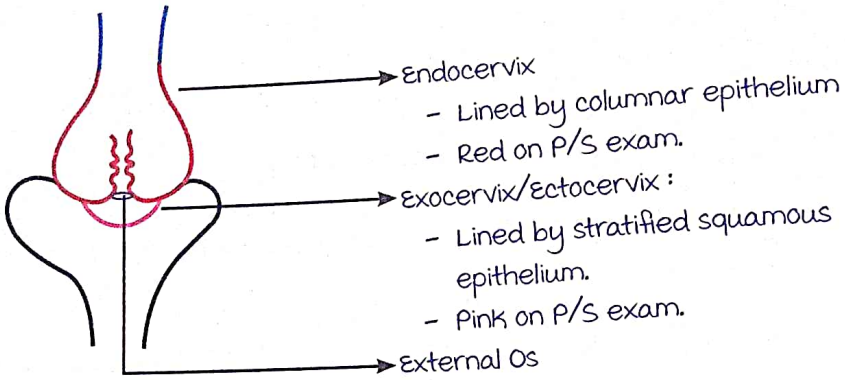
CIN : PART 1

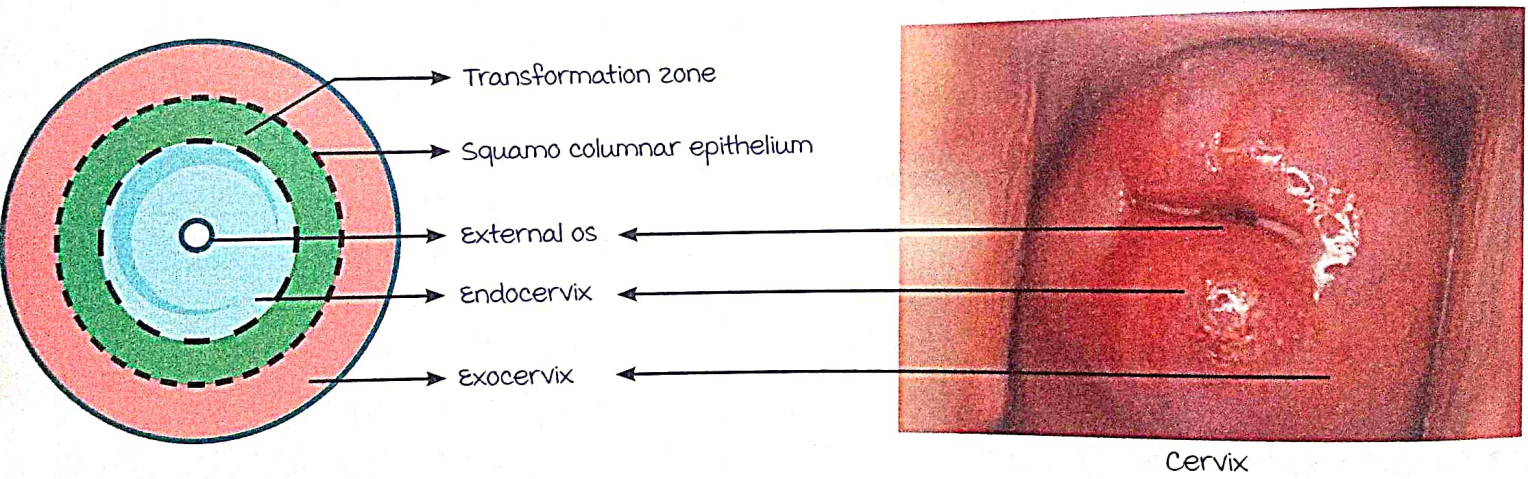
----- Active space -----

00:00:40

Anatomy & Histopathology of Cervix

ANATOMY





Cervix

At puberty :

- Presence of glycogen in vaginal epithelium under the effect of estrogen.
- Presence of Doderlein bacilli.

↓

Acts on glycogen → Lactic acid

↓

Acidic pH of vagina.

• Squamous metaplasia : Squamous epithelium grows over-Columnar epithelium. ----- Active space -----

New squamocolumnar junction

Transformation zone :

- Dynamic area between old & new squamocolumnar junction.
- Position changes d/t
  - Hormonal status.
  - OCPs.
  - Pregnancy
  - menopause
- original position : Located near external Os.

Blockage of columnar epithelial glands

Nabothian cysts/Follicles :

- Physiological.
- Biopsy not required.
- Indicates the original Sc junction.

pplied aspect :

ancer cervix :

- Squamous cell carcinoma : most common variety.
- Transformation zone (TZ) : m/c site.
- Endocervix : m/c site for adenocarcinoma.

ap smear :

Samples :

- Transition zone : Near external Os.
- Endocervix.

HISTOPATHOLOGY

Metaplasia vs Dysplasia :

Metaplasia	Dysplasia (Unorganized metaplasia)
Physiological	Pathological
Not premalignant	Premalignant
Seen in all females	Occurs in presence of risk factors by ca cervix

Characteristics of dysplastic cells :

- ↑ Nucleus : Cytoplasmic ratio.
- ↑ Nuclear pyknosis.

----- Active space -----

Cervical epithelial thickness	Carcinoma
< 1/3rd	CIN 1
1/3 – 2/3	CIN 2
≥ 2/3	CIN 3
Complete thickness without involvement of overlying membrane	Carcinoma in situ (CIS)
Involvement of overlying membrane	Invasive carcinoma of cervix

Applied aspect :

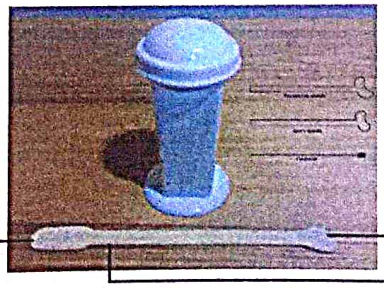
1. CIN 1, 2, or 3 : Dx by biopsy or tissue sample.
2. Pap smear :
   - Cytological test.
   - Identifies dysplasia :
     - Extent cannot be assessed.
   - Inference → Dysplasia +
     → Dysplasia –
     - Abnormal pap smear → Biopsy.
   - Mx is not based on pap smear report.

Pap Smear

00:22:42

Conventional pap smear :

Instruments :
Ayres spatula.
Endocervical brush.



Uses of other end :
Vaginal cytology,
Buccal smear.

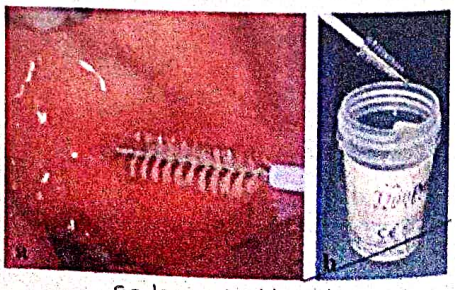
Bifid end

Ayres spatula

endocervical brush

Samples :

3. Ectocervix :
   - From transition zone with bifid end of Ayer's spatula.
   - Near external os.
   - Rotated 360°.
4. Endocervix :
   - Using endocervical brush.



Endocervical brush

Procedure :
- Both smear samples on same slide.
- Fixative : 95% ethyl alcohol ± 5% ether.
- Do not air dry the slide.

----- Active space -----

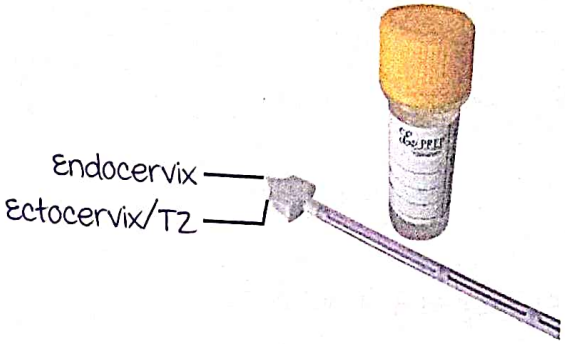
Relative contraindication : Active bleeding.

Notable points :
- P/V exam is not done prior pap smear.
- Best time for pap smear : Periovulatory phase.
- Separate cervico-vaginal swab is taken for HPV DNA testing.

Liquid based cytology :

Instruments : Cervical brush/broom.

Fixation : methanol.



Cervical brush

Advantages :
- Same sample used for pap smear & HPV testing.
- Can be tested during menstruation.
- Less no. of unsatisfactory smears.

Sensitivity : Same for conventional pap smear & liquid based cytology.

Bethesda classification of cytopathology smear :

	Conventional pap	Liquid based cytology
Squamous cells	8K–10K	4K–5K
Columnar cells	10–12	10–12

Risk Factors

00:33:43

- HPV.
- Early coitarche < 18 yrs.
- Early childbirth <20 yrs.
- Multiple sex partners.
- Immunocompromised state.
- Multiparity.
- HIV.
- OCPs : Reversible risk factor → Adenocarcinoma of cervix.
- Smoking → Squamous cell carcinoma.
- Low socio-economic status.

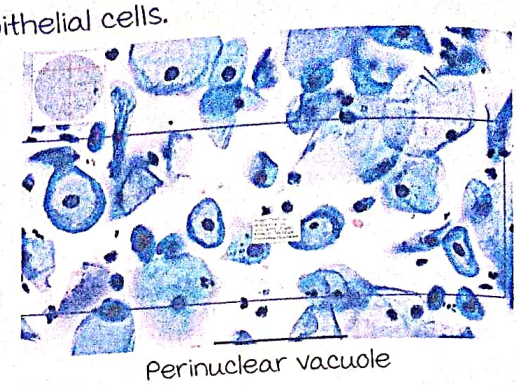
Notable features :
- Cervix ca has no relation to early menarche & late menopause.
- No role of familial inheritance in ca cervix.
- HRT does not ↑ risk of ca cervix.

----- Active space -----

HPV :

Epitheliotropic : Completes its life cycle in the epithelial cells.

HPV → Affects basal & parabasal cells
↓ Completion of life cycle
Superficial cells
↓
Koilocytosis → Perinuclear halo/vacuole



Perinuclear vacuole

Malignant transformation :

E6
↓
Knocks out p53 gene
↓
Immortal cell

E7
↓
Knocks out retinoblastoma gene

Subtypes & functions :

Viral replication : E1, E2 :

- E2 protein → Upregulate E6, E7.

High risk subtypes :

- HPV 16 & HPV 18 (70% cause of ca cervix) :
- m/c : HPV 16 (50%).
- Most specific : HPV 18.
- Others : 31, 33, 35, 52, 58.

High risk HPV subtypes
↓
Cancer

Males :	Females :
• Penis Ca	• Ca Cervix
• Anal Ca	• Ca Vulva
• Oropharyngeal Ca	• Ca Vagina

Low risk subtypes : HPV 6 & 11 →
- Genital warts/Condyloma Acuminata
  - DOC in Pregnant female : Trichloroacetic acid (TCA)
- Laryngeal papilloma in newborns

00:44:24
----- Active space -----

methods :

- screening.
- vaccines.

SCREENING

- HPV infection
  - 90% → Resolved
  - 10% → persists → Dysplasia → CIN (21–30 yrs) → Ca in situ (30–35 yrs) → Ca cervix (35–39 yrs & 60–65 yrs)
- (10–30 yrs)
- universal screening.

WHO goals (2030) :

90% : vaccination coverage for ca cervix by 15 yrs.
70% : screened by 35 yrs → Repeated at 45 yrs.
90% : with CIN or Ca cervix → Receive Rx.

methods :

- most sensitive : HPV DNA testing (85–90%).
- most specific : Pap smear (55% sensitivity).
- Other methods (Recommended by WHO) : VIA – Visual inspection with acetic acid :
  - Not recommended by WHO : VILI – Visual inspection with Lugol's iodine.

Screening protocol :

- Resource rich countries : ACOG protocol.
- Resource limited countries : WHO protocol.

WHO protocol :

Started at 30 yrs → until 50 yrs.

Approaches

See & treat :
- 1° screening test
  - +ve
  - Rx
- HPV DNA testing > VIA

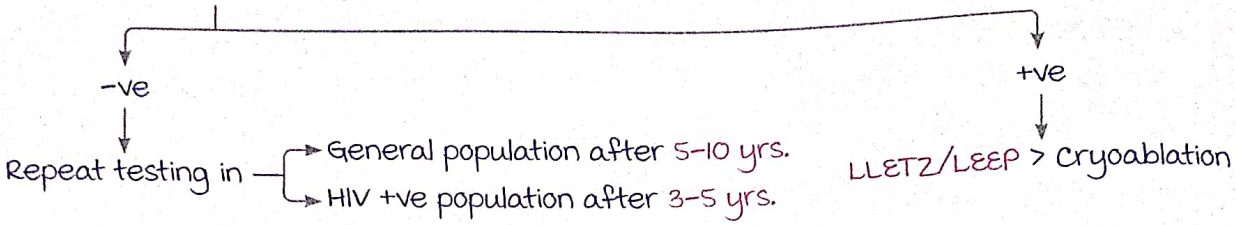
<

See, triage & treat :
- 1° screening
  - +ve
- 2° screening
  - +ve
  - Rx

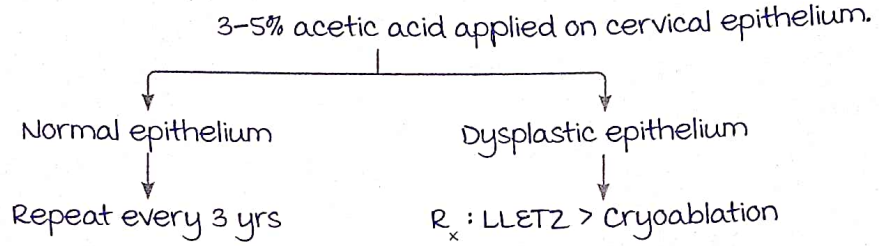
----- Active space -----

See & treat approach with HPV DNA :

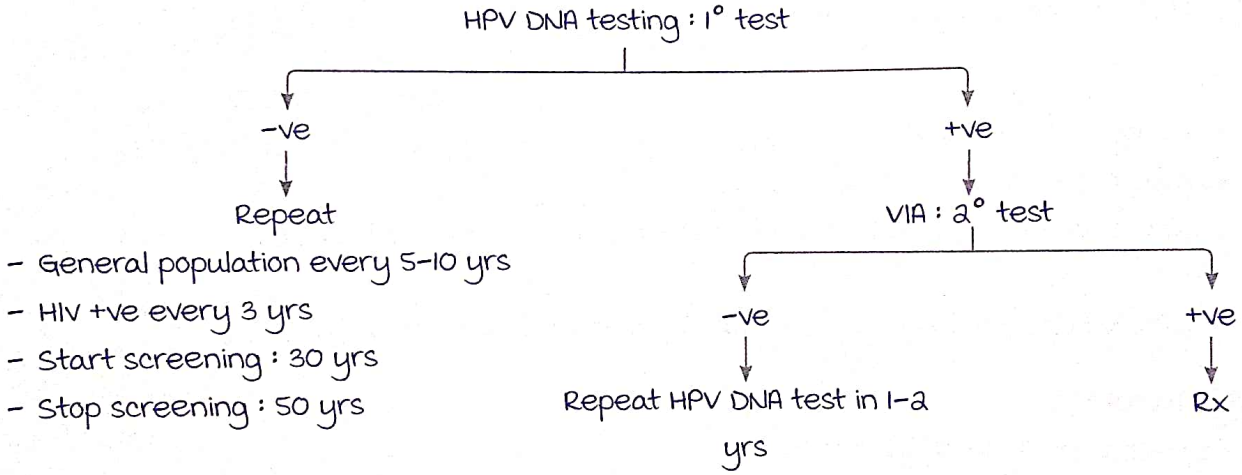
- Start screening at 30 yrs → Until 49 yrs.
- HPV DNA testing : 1° test



See and treat approach with VIA :



See, triage and treat approach :



Management :

LLETZ : Large loop excision of transformation zone/LEEP : loop electroexcision procedure.

Cryosurgery :

Indication : All 4 of the following to be present :

- TZ is visible.
- Entire lesion visible.
- Non - HIV patient.
- Dysplastic area should occupy <2 quadrants of cervix.

----- Active space -----

HPV DNA testing :
- Done at ≥ 30 yrs – Earliest at 25 yrs.
- Testing via : Hybrid capture technique.
- Indicator of high risk HPV infection :
  - Does not tell about subtype of HPV.

Tests for subtype of HPV :
- Onclarity test.
- Cobas test.
- Partial genotyping test.

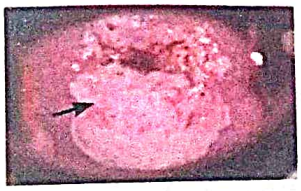
1. HPV test + Pap test : Cotest.
2. Pap test $\xrightarrow{\text{Abnormal}}$ HPV test : Reflex HPV testing.

VIA :
- Apply 3–5% Acetic acid

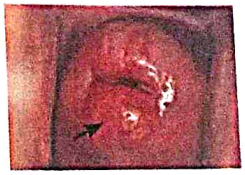
Dysplastic cells → ↑ Nucleus → ↑ Nuclear protein → Coagulate acetic acid → Acetowhite appearance

Inference

White → Dysplastic epithelium



Pink → Normal epithelium



Grey → metaplasia

VILI :

Visual Inspection with Lugol's Iodine

Stained area → Normal epithelium

Unstained area → Dysplastic epithelium

Note : Ectropion
- Endocervix visible on ectocervix.
- M/C in Pregnancy.
- Not premalignant.
- C/o post coital bleeding.

Endocervix – Not visible normally

Ectropion

----- Active space -----

ACOG recommendations :

Screening from 21 yrs :

Pap smear
↓
Repeat 5 yearly
↓
Till 30 yrs
↓ 30-65 yrs
HPV DNA testing + Pap smear : Cotest
↓ Repeat
Every 5 yrs until 65yrs

Notable points :

- Hysterectomy done for benign causes :
  - Stop screening after hysterectomy.
- Female begins sexual life <21 yrs :
  - Screening started from 21 yrs.
- Females between 55-65yrs : ≥ 1 abnormal pap smear
  - Screening continued upto 75 yrs.
- Screening continued even after HPV vaccinations.

HPV VACCINES

All vaccines prepared from L1 capsid protein.

Cervavac :

- Quadrivalent.
- Subtypes : HPV 6, 11, 16, 18.
- Protect against genital warts, ca cervix.
- Manufactured by Serum Institute of India, Pune.

Gardasil :

- Quadrivalent.
- Subtypes : HPV 6, 11, 16, 18.

Cervarix :

- Bivalent.
- Subtype : HPV 16, 18.

Gardasil 9 :

- m/c used HPV vaccine in the US.
- Nonavalent.
- Subtypes : HPV 6, 11, 16, 18, 31, 33, 45, 52, 58.

Age :

- Ideal age : 11-12 yrs.
- Age group : 9-26 yrs.

- High risk females : 27 yrs – 45 yrs.
- Vaccine approved by FDA : Till 45 yrs.

----- Active space -----

WHO SAGE Recommendations :

Goals : 90% of females vaccinated by 15 yrs of age.

Doses :

Age	Doses
9-20 yrs	1/2
≥21 yrs	2
HIV +ve	3

HPV vaccines can be given to :

- HIV +ve.
- Sexually active females.
- Females with HPV.
- Boys (should be given).

Vaccine schedule (old) :

Age	No. of doses	Dose	Schedule
<15 yrs	2	0.5 ml, IM	0, 6 months
15-26 yrs	3	-	0, 2, 6 months

- No need for HPV testing before giving vaccine.

Contraindications : Pregnancy.

Side effect : Syncope.

----- Active space -----

CIN : PART 2

CIN : Cervical intraepithelial neoplasia.

Features of CIN

00:01:20

CIN based on dysplasia :

Pap smear	CIN	Risk of cancer cervix
LSIL	CIN I	1%
HSIL	CIN 2	5%
	CIN 3	12 - 40%

- CIN is diagnosed on biopsy, not on pap smear.

WHO guidelines on Mx :

In limited resource countries :

- Screening → HPV DNA ± VIA → Mx.
- Biopsy is not mandatory.

In resource rich countries :

Pap smear —Abnormal→ Biopsy → CIN I/2/3 → Mx.

PAP SMEAR

Inference of Pap smear :

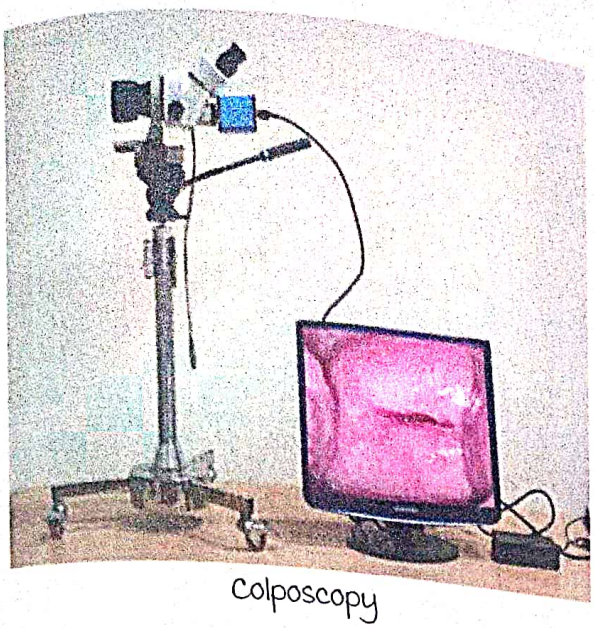
- Normal.
- Infection → Mx : Antibiotics.
- Low squamous intraepithelial lesion (LSIL).
- High squamous intraepithelial lesion (HSIL).
- Atypical squamous cells of unknown significance (ASCUS).
- Atypical squamous cells where HSIL cannot be ruled out (ASC-H).
- Atypical glandular cells of unknown significance (AGCUS).

Dx of infection in pap smear :

- Bacterial vaginosis.
- Candida.
- Trichomonas.
- HSV, HPV.

Infections not detected by pap smear :

- Chlamydia.
- Gonorrhea.



Colposcopy

Pap smear : Screening
↓ Abnormal
Per speculum examination
→ Growth + → Punch biopsy
→ No visible Abnormality → Colposcopic guided biopsy

----- Active space -----

Mx based on pap smear reports :

- HSIL, ASC-H } Any age
- LSIL ≥ 25yrs → Colposcopy + Endocervical curettage.
- ASCUS — Age → < 25 yrs : Repeat pap in 12 months.
  → > 25 yrs : HPV DNA testing / Reflex DNA testing.
- LSIL in < 25 yrs: Repeat pap in 12 months.
- AGCUS → Adeno carcinoma of endocervix.
  → Endometrial carcinoma spread to cervix.
  1. Colposcopic biopsy ⊕
  2. Endocervical curettage ⊕
  3. Endometrial biopsy.

COLPOSCOPY

- magnifying instrument : × 30 times.
- Focal length : 30.
- Visualizes : Exocervix, Transformation zone, External os.
  - Does not visualize : Endocervix.
- UPT done before colposcopy if needed.
- OPD procedure.

Samples :

1. Site : Rough or pale areas.
2. Apply 3-5% acetic acid on cervix → Biopsy taken from acetowhite areas.
3. Green filter turned on → Biopsy from abnormal blood vessels. → Reticular, Mosaic, Punctate

----- Active space -----

Inference :

- CIN 1/2/3.
- Microinvasive cancers.
- Unsatisfactory.
  - Incomplete visualization of transition zone.

Management of CIN

00:22:50

CIN 1 :

Follow up × 2-yrs

- Resolves
- Does not resolve → Cryoablation

CIN 2/3 :

LLETZ or LEEP

- m_x of choice for all age & parity.

Note : Hysterectomy is not done for CIN.

CRYOABLATION

- CO_2 / N_2O passed at very low temperature.
- With cryogen : Applied in areas of dysplasia (Acetowhite areas).

Principle :

- CO_2 / N_2O —(At very low temp)→ Freezing & intracellular water in dysplastic areas → Destruction of dysplastic epithelium
- Freeze (3 mins) → Thaw (5 mins) → Freeze (3 mins)
- Destruction upto 3 – 5 mm of epithelium.

Side Effect : Persistent water discharge.

Disadvantage : No tissue specimen is obtained

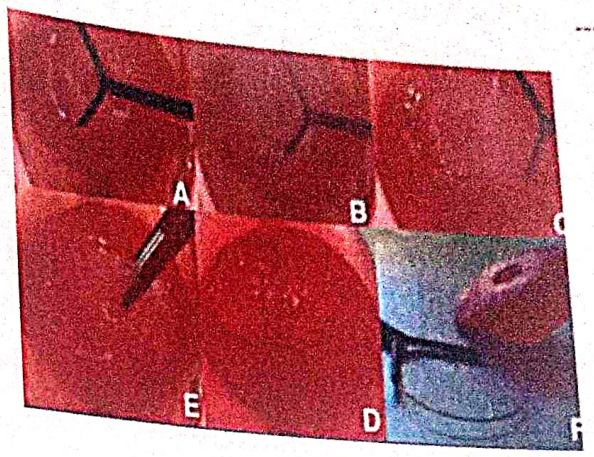
Prerequisites :

Done in CIN 1 : All of the following criteria to be fulfilled.

- Entire TZ should be visible.
- Lesion should be completely visible.
- Not be HIV +ve.
- Lesion not > 2 Quadrants of cervix.

LLETZ/LEEP

- Total dysplastic area is removed.
  - Current passed into Instrument
  - Cut & coagulate at the same time



LLETZ

Advantage :

- OPD procedure.
- Short procedure : 10–15 mins.
- Tissues removed until 10 mm depth.
- TOC for CIN 2 & 3 for any age & parity.
  - Tissue sample can be sent for HPE.

Disadvantage : Bleeding (minimal)

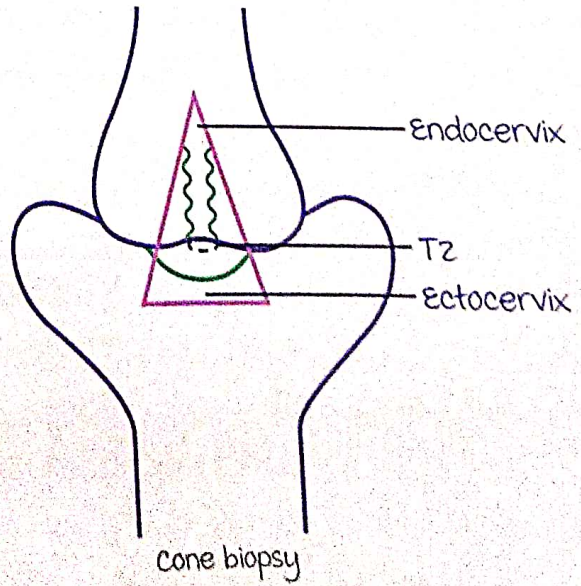
SPECIAL SCENARIO

1. Endocervical curettage → Cone biopsy → Adenocarcinoma in situ → Hysterectomy
   positive

2. PAP smear : HSIL → Colposcopy & biopsy : CIN I or Normal.
   ↓Discrepancy in reports
   Cone biopsy

CONE BIOPSY / CONISATION

- Cone shaped area of cervix is cut.
- To include :
  - Exo/ecto cervix.
  - Transformational zone.
  - Endocervix.
- Done in OT under GA.



Cone biopsy

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Indication :

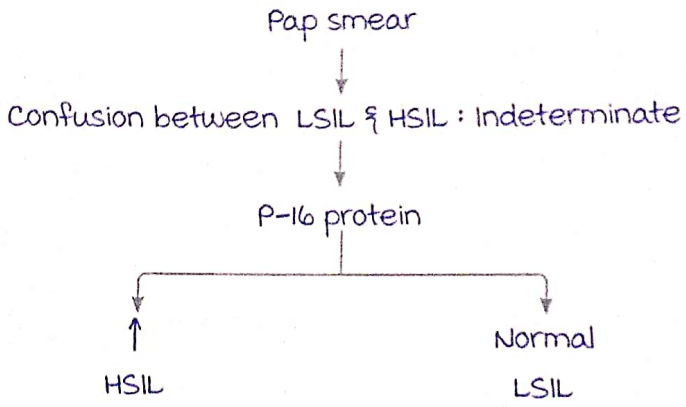
Diagnostic :

- Suspecting adenocarcinoma/adenocarcinoma in situ.
- Poorly visualised TZ on colposcopy.
- Discrepancy in pap smear & colposcopy reports.

Therapeutic :

- mx of Ca cervix in young females.
  - Stage IA_1 : microinvasive Ca
- Evidence of microinvasion.

P-16 PROTEIN :



CANCER CERVIX

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General Characteristics

00:00:23

Types of cancer :

Squamous cell cervix : m/c type of cancer cervix.
- Large cell non-Keratinising type (m/c type).
- Large cell Keratinising type.
- Small cell (poor prognosis).

- m/c association: HPV 16.

Adenocarcinoma : 2nd m/c :
- m/c in young female.
- Risk factor : OCP.
- m/c association : HPV 18.
- m/c site : Endocervix.

Features :
- m/c site : Transitional zone.
- m/c age : 35-39 yrs & 60-64 yrs.

Symptoms :
- Irregular bleeding : m/c symptom.
- Post coital bleeding : Most specific symptom.
- Dirty vaginal discharge.
- Pyometra (m/c Ca causing pyometra : Ca cervix).
- Post menopausal spread.
- Dyspnea, cough, hemoptysis, chest pain : Lung metastasis.
- Sciatica, Hematuria, Lymphedema : Pelvic side wall involvement.
- m/c cause of death : Renal failure.

Route of spread :
- Direct > Lymphatic > Hematogenous :
  - m/c site : Lungs (Hematogenous route).

Drainage :
- Hypogastric (Internal iliac) nodes.
- Obturator node.

----- Active space -----

- Paracervical node : Sentinel node.
- External iliac nodes.
- Cervix does not drain into superficial lymph nodes :
  - Not included in radiotherapy.

Management :

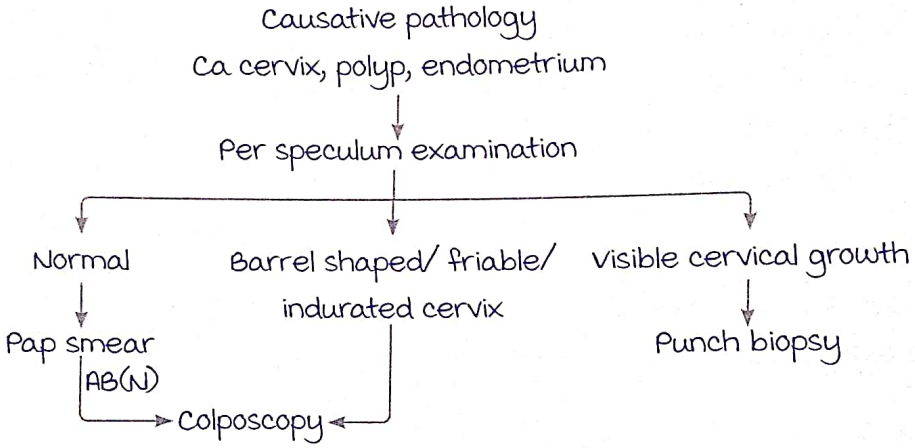
- External Beam Radiotherapy (EBRT) : Pelvic lymph nodes involvement.
- Extended EBRT : EBRT + Paraaortic lymph nodes involvement.

Sites involved in cancer cervix :

Parametrium, pelvic side walls, bladder, bowel, lungs, Kidney.

Post Coital bleeding presentation :

Algorithm :



EXAMINATION

General examination :

- Thin built, cachexic.
- Low socio-economic status.

P/A examination :

- Abdomino pelvic lump d/t pyometra (Collection of pus in the uterus).

P/S examination :

- Growth : Ulcerative/Fungating mass.
- Friable cervix: Bleeds on touch.
- Hard & nodular cervix.

Per rectal/Per vaginorectal examination :

- Examination of parametrium and utero sacral ligaments :
  - Thickening & nodularity.

Investigations

00:14:38 ----- Active space -----

Prognostic markers : Staging > Lymph nodes.
Diagnostic : Cervical biopsy.
For microinvasive cancer on LEEP → Cone biopsy.

Note : Involvement of ovaries (<1%).

- Hysterectomy for Ca cervix in young females.
  ↓
  Ovaries preserved : To avoid pre-mature menopausal symptoms.

STAGING

- Clinical staging + Radiological investigation :
  - USG : Presence of pyometra.
  - CT scan : Lymph node status.
  - MRI : Involvement of parametrium.
  - PET CT : Lymph nodes involvement.
  - IVP : Kidney & ureter involvement.
  - Cystoscopy : Bladder involvement.
  - Sigmoidoscopy : Rectum involvement.
  - Chest X-ray : Lung involvement.
- Best investigation for :
  - Involvement of parametrium : MRI.
  - LN involvement : PET-CT > CT scan.

FIGO staging for cancer cervix :

Stage		Features
IA	Microscopic/microinvasive cancer ≤5 mm deep.	
	IA_1	≤3 mm deep.
	IA_2	4-5 mm deep.
IB	Macroinvasive cancer >5 mm deep.	
	IB_1	Size ≤ 2 cms
	IB_2	Size ≤ 4 cms
	IB_3	Size > 4 cms
II	Involvement of upper 2/3rd of vagina.	
	IIA	Parametrium not involved.
	IIA_1	Size : ≤ 4 cms
	IIA_2	Size: > 4 cm

----- Active space -----

Stage	Features
IIB	Parametrium involved.
III	Involvement of lower 1/3rd of vagina.
IIIA	Pelvic side wall not involved.
IIIB	Pelvic side wall involved : Hydroureter/Hydronephrosis (Ureter, Kidney involvement).
IIIC	Lymph nodes involvement.
$IIIC_1$	Pelvic lymph nodes involvement.
$IIIC_2$	Para aortic lymph nodes involvement.
IV	metastasis
IVA	Regional metastasis : Bladder/Bowel.
IVB	Distant metastasis/Superficial inguinal lymph node involvement.

- Spread of cancer cervix to uterus : Does not change staging.
- Bullous edema of bladder : Not a metastasis.
- metastasis of bladder : Histopathological evidence +.

Management

00:28:35

Surgery :

Indications :

- Uptil stage IIa.
- Size of tumor < 4 cm.
- Preferred in early stage (except IB_3).

Radio-therapy < Chemoradiation :

- Can be done in all stages.
- Preferred when S_x is contraindicated.
- Radiosensitizer : Cisplatin, 5FU.

Surgery :

Stage	Complete family		Incomplete family (Ovaries preserved)
	Hysterectomy	LN dissection	
IA_1 (No lymphovascular invasion)	Type I	No	Conization (Cone biopsy)
IA_2	Type 2 / Wertheim	Pelvic lymph node	Radical trachelectomy + Pelvic LN dissection
IB_1	Type 3	Pelvic lymph node	Type 3 Hysterectomy + Pelvic lymphadenectomy
IB_2			
IIA_1			

Radical trachelectomy :

- Cervix removed.
- Uterus stitched to vagina.
- Followed by abdominal cerclage (Internal as injured during removal of cervix).
- Indication :
  - Done for young females who want future child bearing.
  - Size < 2 cm.
- mode of delivery post procedure on conception : LSCS.

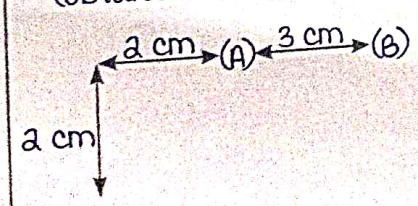
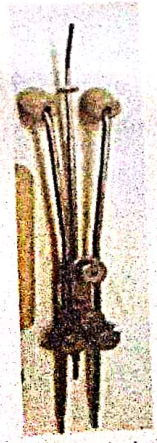
----- Active space -----

Chemoradiation :

- Indication : Stage IB_3, IIA_a – IV.
- Radiosensitizer :
  - ↑ Sensitivity to Radiotherapy (RT).
  - Given prior to RT.
  - Cisplatin > 5FU.
- Side effects :
  - Vaginal fibrosis (Difficult intercourse).
  - Ovarian failure : Ovary is most radiosensitive organ.

} RT not preferred in early stages

Types :

	Brachytherapy	Teletherapy
Source of radiation	• Inside body • Intracavitary therapy • Using Colpostat/Tandem	• Outside the body • External beam radiotherapy (EBRT)
Isotope	Iridium 192	Cesium
Indication	Following teletherapy	Given 1st to shrink size of tumour
other features	• Point A : 2 cm above & lateral to external os (Parametrial node). • Point B : 3 cm away from point A (obturator nodes).   Colpostat	50 Gy : – 5 times a week. – 2 Gy per sitting.

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Post - surgery :

High risk :

- Margin +ve.
- Parametrium +ve.
- LN metastasis +ve.

↓

Post-op chemo radiation.

Intermediate risk :

- Large size of tumour.
- Lymphovascular involvement.
- Deep stromal invasion.

} Any 2 +ve

↓

Post-op radiotherapy.

Low risk :

↓

No post-op mx.

OVARIAN CANCER : PART 1

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Ovarian cysts & Ovarian cancer

00:00:44

OVARIAN CYSTS

Functional ovarian cyst :

- m/c type, & Occurs d/t hormonal disturbances.
- Temporary : m/c resolve spontaneously (Specific treatment not required).

Types :

1. Follicular cyst (m/c) : size of follicle ≥ 3cm.
2. Corpus Luteal cyst : m/c cyst to rupture.
3. Theca Lutein cyst : D/t ↑ HCG
   - Seen in : molar pregnancy, Twin pregnancy, (Infertility R_x Clomiphene, HMG)

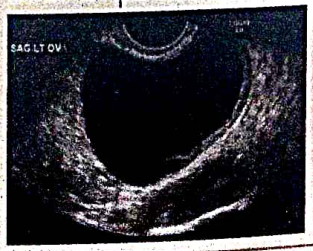
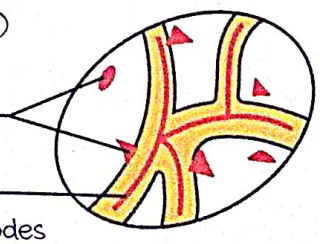
FEATURES OF OVARIAN CANCERS

Rare, therefore screening not done.

Presentation :

- Vague symptoms (Nausea/Vomiting, Loss of appetite)
- Adnexal masses (IOC : TVS)

Benign V/s malignant adnexal masses

	Benign adnexal mass	malignant adnexal mass
Prevalence	m/c in reproductive age group	Extremes of age : Pre-pubertal/menopausal
Pain	⊕ (d/t Inflammation)	⊖
History	-	Short history with rapid progression & weight loss
Occurence	Unilateral	Bilateral
Consistency	Cystic	Solid
Tenderness	⊕	⊖
Ultrasound features	• U/L • Anechoic • Unilocular • Solid component ⊖ 	• B/L, > 10 cm size • Variable consistency (Solid components) • Thick septa + – With papillary outgrowths (Excrescenes). – Highly vascular. • Evidence of ascites, enlarged lymph nodes & matted bowel loops. 

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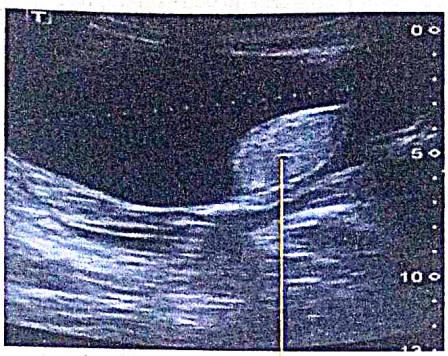
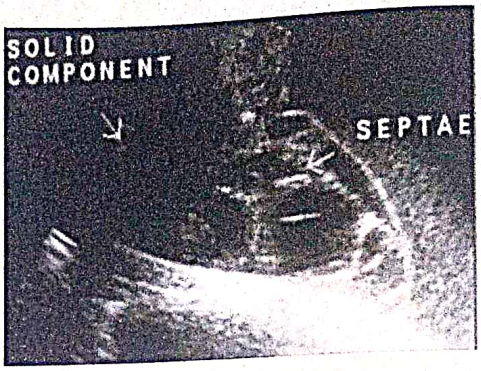
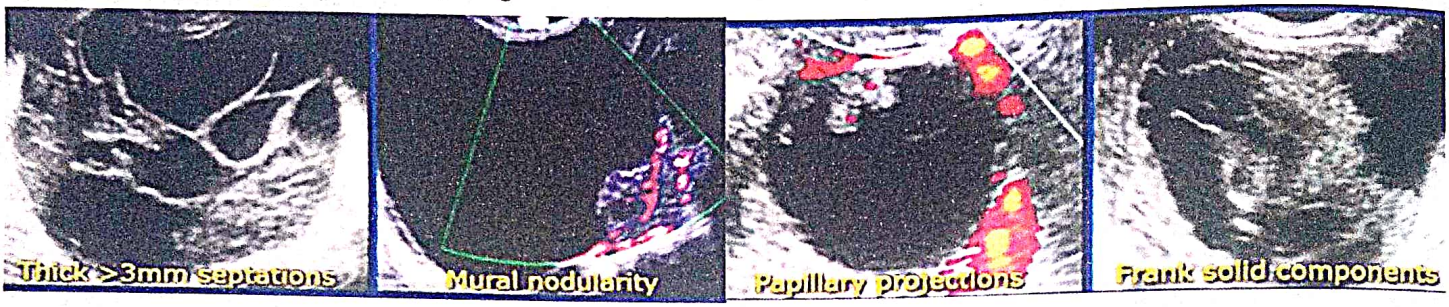




Papillary excrescenes



Features suggestive of Ovarian malignancy on USG

MANAGEMENT OF OVARIAN CYST

- Reproductive age group
  - Cyst size
    - 3–5 cm → Wait and watch
    - 5–7 cm → Follow up with serial USG
    - > 7 cm → Surgery (High chances surgery of torsion or rupture)
- Extremes of age
  - Post-menopausal → CA 125 ≥ 35 IU → Surgery after further investigations
  - Pre-pubertal → If any raised : AFP, HCG, LDH → Surgery after investigations

Note:

1. % of malignant transformation of ovarian mass → Post menopausal : 30%; Pre menopausal : 7%
2. CA – 125 :
   - Limited value in reproductive age group :
     - ↑ In benign conditions (Fibroid, PID, Genital TB).
     - 200 IU : Significant.

OVARIAN PATHOLOGIES IN PREGNANCY

m/c
- Benign ovarian tumor cyst.
- To undergo torsion.

} Dermoid cyst (mature cystic teratoma)

- Time for torsion : End of 1st trimester/ Puerperium.
- Ovarian cancer : Dysgerminoma.

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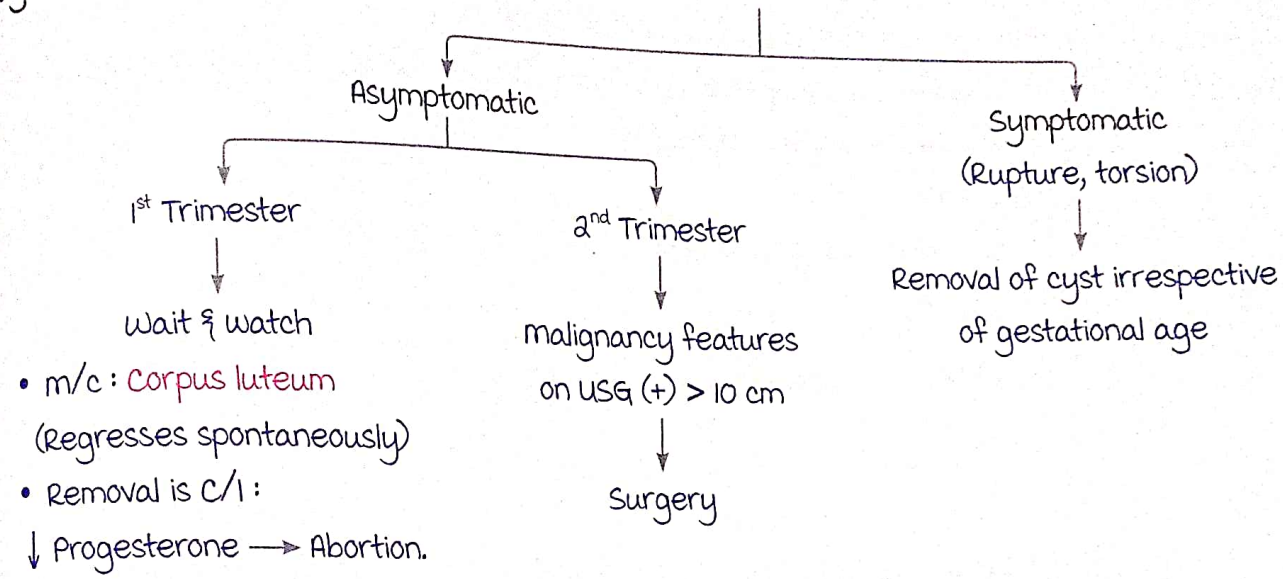
management :



- m/c : Corpus luteum (Regresses spontaneously)
- Removal is C/I :

↓ Progesterone ⟶ Abortion.

Factors Related to Ovarian Cancer

00:25:03

Risk Factors :

	Proven risk factors	Controversial risk factors
Excessive estrogen	• Early menarche. • Late menopause. • Obesity. • Endometriosis.	• PCOS • HRT
Excessive ovulation (Theory of incessant ovulation)	Nulliparity	Infertility & ovulation inducing drugs
Genetic syndrome	• Lynch syndrome. • BRCA 1, BRCA 2.	
Exposure to carcinogens	Asbestos	• Talc. • Smoking (mucinous adenocarcinoma)

Protective Factors :

Related to estrogen	Related to ovulation	Surgeries
Physical exercise	• Anovulation • multiparity • OCP • Breast feeding	• Hysterectomy • Tubal ligation • Salpingectomy

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GENETIC SYNDROMES

1. Lynch syndrome :
   - m/c gene : MLH_1 / MSH_2
   - m/c associated cancer : Colorectal cancer > Endometrial cancer (60–70% risk)
   - Risk of ovarian cancer : 20%
2. BRCA 1 :
   - Maximum risk of ovarian cancer (40%)
   - Prevention : TAH + BSO after completion of family (35–40yrs of age)
   - Risk of endometrial cancer ⊕
3. BRCA 2 :
   - 15% risk of ovarian cancer.

Hereditary vs sporadic ovarian cancer :

- Hereditary :
  - Affects younger age group (50 years).
  - 5–10% of all ovarian cancers.
- Sporadic :
  - m/c in 60–70 yrs.

Screening :

1st degree relative

- Hereditary ovarian cancer/$BRCA_1$ mutation ⊕ → Annual screening : TVS + CA 125 (35–40yrs)
- Sporadic ovarian cancer → Reassurance

WHO classification & Epithelial Ovarian Tumors

00:41:50

WHO classification of ovarian tumors :

- Epithelial ovarian tumors : m/c (90%)
- Germ cell tumor : 2nd m/c (5–8%)
- Sex cord stromal : 3rd m/c (3–5%)
- metastatic tumors

m/c
- Ovarian tumor
  - Overall : Serous cystadenoma.
  - Reproductive age : Dermoid cyst (mature cystic teratoma).
- Ovarian cancer : Serous cystadenocarcinoma

EPITHELIAL OVARIAN TUMORS

----- Active space -----

Types :

- Serous tumors (m/c) : → Benign : Serous cystadenoma.
  → Malignant : Serous cystadenocarcinoma.
- mucinous tumors.
- Brenner's tumor.
- Endometroid tumor.
- Clear cell tumor.

Features :

- m/c age : 60 yrs.
- B/L (m/c).
- Worst prognosis : d/t non-specific symptoms (Nausea, Vomiting, IBS, weight loss).
- HPE : Psammoma body.
- mutations : → Low grade tumors : KRAS, pTEN mutation.
  → High grade tumors : p53 mutation.

Serous vs Mucinous tumors :

	Serous tumors	mucinous tumors
% Benign	60% (Serous cystadenoma)	80% (mucinous cystadenoma)
Occurence	mostly bilateral	mostly unilateral
Associated mutation	BRCA-1, BRCA-2, p53 mutation	KRAS mutation
Smoking	Not a risk factor	Risk factor (+)
malignant counterpart	Serous cystadenocarcinoma	mucinous cystadenocarcinoma
Tumor marker	CA 125	CEA, CA 19-9
HPE	Psammoma bodies	-
Gross finding	Uniloculated cyst filled with clear/serous fluid	Honeycomb appearance : multiloculated cyst filled with mucinous material.
Microscopy findings	Resembles fallopian tube lining	Resembles endocervix lining
Associated with	-	Pseudomyxoma Peritonei

----- Active space -----

Pseudomyxoma Peritonei :

- Peritoneum filled with mucinous material
- Causes :
  - Appendix cancer : m/c.
  - Mucinous ovarian tumor (metastatic).
  - Mucocele of appendix.
- Bad prognosis : d/t high recurrence rate.

Brenner's tumor :

Mnemonic : We BUST

Walthard cell nest + Coffee bean nuclei on HPE

Benign tumor

Unilateral (100%)

Solid tumor

Transitional epithelium lining

Endometrioid v/s Clear cell tumor :

	Endometrioid Tumor	Clear cell tumor (malignant)
HPE	Resembles endometrial glands	Hobnail cells
Associations	a. Endometriosis. b. Endometrial cancer.	a. Endometriosis. b. In-utero DES exposure.

Note :

1. Coffee bean nuclei (Central grooving ⊕) → Brenner's tumor. / Granulosa cell tumor.
2. m/c ovarian tumor associated with
   a. Endometriosis : Clear cell tumor > Endometrioid tumor.
   b. Endometrial cancer : Endometrioid tumor > Granulosa cell tumor.
3. Solid Tumors : Brenner's & Fibroma.

Germ Cell Tumors (GCT)

00:59:45

Varieties :

- Teratoma → Benign : Mature cystic teratoma (Dermoid cyst)
  → Malignant : Immature teratoma
  → Monoclonal : Struma ovarii
- Dysgerminoma
- Yolk sac tumor (endodermal sinus tumor).
- Embryonal cancer.
- Choriocarcinoma.
- Mixed type

Features :

- m/c : Young girls (10–30 yrs).
- U/L (m/c).
- Better prognosis : Diagnosed early.

Tumor markers :

GCT	Tumor marker	
	Seen	Never seen
Dysgerminoma	• LDH • hCG • PLAP (Placental alkaline phosphatase)	AFP
Yolk sac tumor	• AFP • LDH • α_1 antitrypsin	hCG
Embryonal cancer	• AFP • hCG	LDH
Choriocarcinoma	hCG	AFP, LDH
Teratoma	No tumor marker	

----- Active space -----

Occurrence :

- m/c GCT : Dermoid cyst > Dysgerminoma.
- m/c germ cell cancer : Immature cystic teratoma.
- Dysgenetic gonads :
  - m/c ovarian tumor : Gonadoblastoma.
  - m/c ovarian cancer : Dysgerminoma.

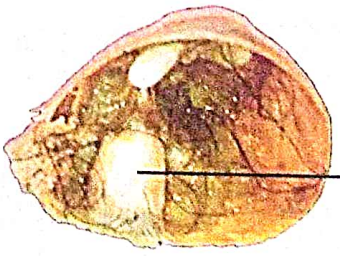
Prognosis :

- Best prognosis : Dysgerminoma.
- Worst prognosis : Yolk sac tumor (Progresses rapidly, presents with acute abdomen)

DERMOID CYST

Features :

- m/c ovarian tumor → In reproductive age / In pregnancy / To undergo torsion
- m/c unilateral : Bilateral in 10% cases.
- All 3 components of germ cell layers ⊕ (m/c : Ectodermal).
- Mostly benign.



Rokitansky protuberance

Gross specimen of Dermoid cyst showing tooth & hair

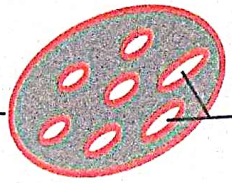
Malignancy :

- Risk : 0.2–2%
- m/c type : Squamous cell carcinoma.
- m/c site : Rokitansky protuberance.
  (Raised prominence giving rise to various components)

----- Active space -----

USG findings :

1. Tip of iceberg sign.
2. Dot & dash appearance.



Sebacious material + Hair (White)

Management :

- Cystectomy (↑ Risk of torsion).
- Oophorectomy : Family completed.

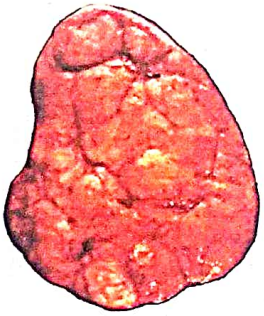
DYSGERMINOMA

Features :

- 2nd m/c GCT.
- U/L : m/c (15-20%; B/L).
- Highest risk of bilaterality.
- Best prognosis.
- m/c ovarian cancer → In dysgenetic gonads. / In pregnancy.
- Only radiosensitive ovarian tumor

Gross appearance :

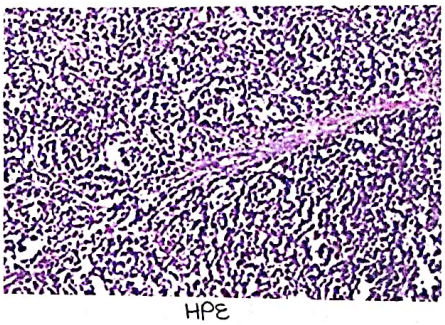
Fleshy, solid, lobulated, tan colour.



Gross appearance of Dysgerminoma

HPE :

- Nest of cells separated by fibrous septa.
- Lymphocyte infiltration in septa ⊕



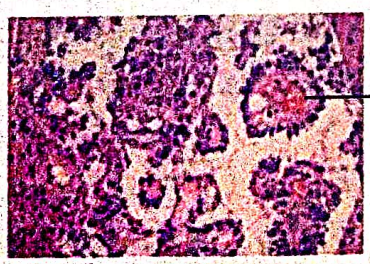
HPE

YOLK SAC TUMOR

Features :

- most malignant GCT.
- Worst prognosis : Rapidly progressing tumor.
- Acute abdomen +/-
- 100% unilateral

HPE : Schiller Duval body/Glomeruloid body



Central capillaries surrounded by tumor cells } Contained within cystic spaces

OVARIAN CANCER : PART 2

----- Active space -----

Sex Cord Stromal Tumors & Metastatic Ovarian Tumor

00:00:10

SEX CORD STROMAL TUMORS

- Sex cord tumors
  - Estrogen producing (Feminizing tumors) :
    - Granulosa cell tumor.
  - Androgen producing (Virilizing tumors) :
    - Sertoli-Leydig cell tumor.
    - Leydig cell tumor.
    - Hilus cell tumor. (Reinke's crystals seen on HPE)
- Stromal tumors
  - Fibroma.
  - Thecoma.
  - Fibrothecoma.

FEATURES

- m/c affects perimenopausal women.
- Unilateral.
- ↑ Production of sex hormones → Specific symptoms → Early diagnosis.

Note :

Best prognosis : Sex cord stromal tumor > GCT > Epithelial cell tumor.

Granulosa cell tumor :

Pathophysiology

- Origin : Granulosa cells
  - produce :
    - Estrogen
    - Inhibin

    } Tumor markers
  - Antimullerian hormone (At puberty)
- FOX L-2 gene

Presentation :

- Excessive bleeding
  - Atypical uterine bleeding (Perimenopausal).
  - Post-menopausal bleeding.
- Precocious puberty.
- Endometrial cancer : Endometrial sampling mandatory in all cases.

Note :

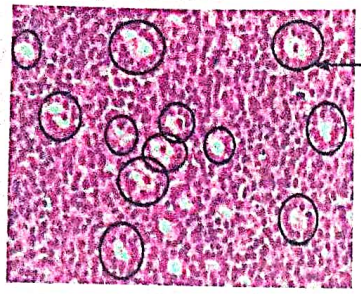
Ovarian cancers a/w endometrial cancer :

a. Endometroid tumor

b. Granulosa cell tumor.

----- Active space -----

HPE : Call Exner body



Central acidophillic material surrounded by follicle like cells + Coffee bean nuclei.

Fibroma :

Features :

- U/L, benign, solid tumor.
- HPE : Well differentiated fibroblasts.
- Associated with meig's syndrome.

meig's vs. Pseudo-meig's syndrome :

meig's syndrome	Pseudo meig's syndrome
• Fibroma/Thecoma/Brenner's tumor/ Granulosa cell + Ascites + Pleural effusion. • Removal of tumor → Resolution of ascites and pleural effusion.	Any other ovarian tumor (mucinous cystadenoma) or any other condition (fibroid) + Ascites + Pleural effusion.

METASTATIC TUMORS OF THE OVARY

- Stomach (m/c)
- GIT
- Breast carcinoma

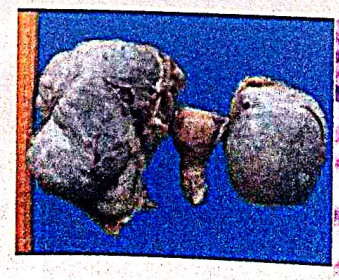
→ Retrograde lymphatics/ Transcoelomic spread → Ovary

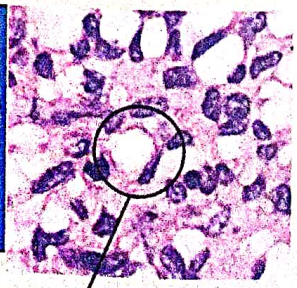
Krukenberg tumor :

Stomach cancer (Pyloric end) → Retrograde lymphatics → metastatic ovarian tumor

Features :

- 80% : B/L
- Symmetric enlargement of ovaries.
- Shape & capsule of ovary : Retained.
- Mobile + waxy consistency.





HPE :

- Signet ring cells :
  - Mucin filled cytoplasm pushes nucleus to one side.

Management of Ovaian Cancer

00:13:12 ----- Active space -----

FIGO STAGING

Stage 1	Limited to ovary.		
	1A	One ovary involved.	Intact capsule
	1B	Both ovaries involved.	
	1C	Capsule ruptured : • $1C_1$: Intraoperative rupture. • $1C_2$: Preoperative rupture. • $1C_3$: Malignant ascites.	
Stage 2	Spreads to other pelvic organs.		
	2A	Fallopian tube or uterus.	
	2B	Bladder, rectum.	
Stage 3	Spreads beyond pelvis into abdomen.		
	3A	• 3A 1 : Retroperitoneal lymph nodes involved. • 3A 2 : Microscopic involvement above pelvic brim.	
	3B	Macroscopic involvement above pelvic brim with size of implant ≤ 2 cm.	± Extension to liver & spleen capsule or surface.
	3C	Macroscopic involvement above pelvic brim with size of implant ≥ 2 cm.	
Stage 4	Distant metastasis.		
	4A	Malignant pleural effusion.	
	4B	Liver/ spleen parenchyma involved OR Inguinal lymph node involved.	

Note : Involvement of bladder & rectum → Ovarian cancer : 2B
→ Other gynaecological cancers : 4A

STEPS OF SURGICAL STAGING

1	Midline incision
2	Ascites → (+) → Ascitic fluid sample : Sent for cytology. → (−) → Saline wash done & sent for cytology.
3	Inspection + palpation of all abdominal organs.
4	Multiple peritoneal biopsies taken.
5	Type I TAH + BSO.
6	Infracolic omentectomy.
7	Pelvic & para-aortic lymph node dissection —f/b→ Closure

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MANAGEMENT

Surgery :

Surgery of choice : TAH (Total abdominal hysterectomy) + BSO (Bilateral salpingo-oophorectomy).

Indications for other surgeries :

1. Conservative Sx : U/L salpingo-oophorectomy (without hysterectomy).
   - GCT : 10-30 yrs of age.
   - Sex cord tumors : Young females.
   - Stage IA epithelial tumor + family not complete.
2. Debulking Sx : Advanced stages (Stages 3, 4).

Post-operative chemotherapy :

	Epithelial cell tumor	Germ cell tumor	Sex cord stromal tumor
Chemotherapeutic agents	6 cycles : IV + Intraperitoneal. Paclitaxel + Carboplatin/Cisplatin	B : Bleomycin E : Etoposide P : Cisplatin	B : Bleomycin E : Etoposide P : Cisplatin
Indications	All stages except stage I A and I B.	All stages except Dysgerminoma stage I (Best prognosis).	Chemotherapy needed only in stage 3 and 4.

Note :

Sertoli-Leydig cell tumor : Androgen secreting ovarian tumor.

- Hirsutism
- Acne
- Baldness
- Enlarged clitoris (Virilization)

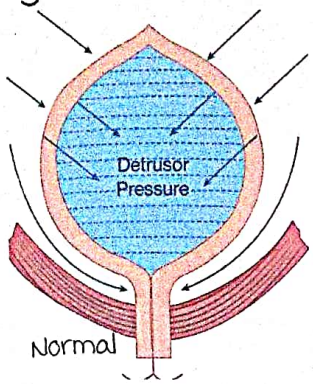
GYNAECOLOGY : MISCELLANEOUS TOPICS

----- Active space -----

Stress Urinary Incontinence (SUI)

00:01:15

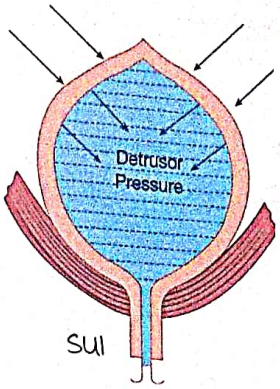
Pathophysiology :



↑Abdominal pressure

- Pelvic floor muscle contracts.
- Intraurethral pressure ↑.

maintains continence



Descent of bladder neck.

↑Intra-abdominal pressure

Failure of urethra to close.

Involuntary passage of urine

Definition :

Involuntary passage of urine on ↑ in intra-abdominal pressure.

E.g. : Coughing, laughing, sneezing.

SUI

Tests for SUI :

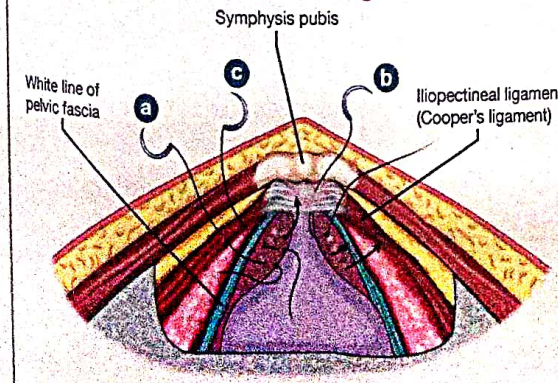
Q tip test.

Management :

Suspension of bladder neck or urethra.

Bursch colposuspension :

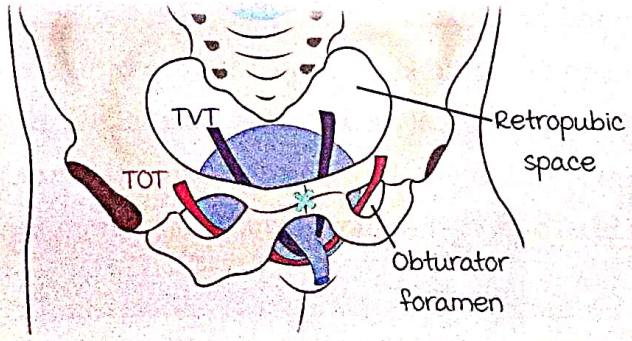
- Abdominal route Sx.
- Proximal urethra & bladder neck suspended to iliopectineal/Cooper's ligament.



- Hospital stay necessary.
- Good outcome : Gold standard.

Tension-free transobturator tape (TOT) / **Tension-free transvaginal tape (TVT)**

- Day care Sx.
- Vaginal route Sx.
- middle urethra is suspended.



TOT :

- Tape passed via obturator foramen.
- m/c Sx (TOT > TVT).

TVT :

- Tape passed via retropubic space.
- ↑ risk of injury to bladder & venous plexus.

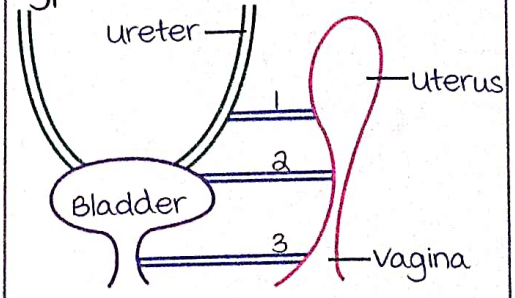
----- Active space ----- **Urinary Fistula** 00:13:06

Complaints :

1. Constant dribbling of urine from vagina + Normal micturition : Ureterovaginal fistula.
2. Constant dribbling of urine from vagina + No normal micturition : Vesicovaginal fistula.
3. Urine passed from urethra and vagina during micturition : Urethrovaginal fistula.

URINARY FISTULA

Types :



1 : Ureterovaginal fistula.
2 : Vesicoivaginal fistula.
3 : Urethrovaginal fistula.

Collection of urine sample : Suprapubic catheterization.

Menouria : Hematuria during menstruation.

- Endometriosis of bladder.
- Vesicouterine fistula.

Tests :

methylene blue 3 swab test :

- IOC for vesicovaginal fistula.
- 3 swabs placed at vagina.
  a. Upper part : Corresponding to ureterovaginal fistula.
  b. middle part : Corresponding to vesicovaginal fistula.
  c. Lower part : Corresponding to urethrovaginal fistula.
- methylene blue dye injected into bladder & urethra by Foley's catheter.

Observation	Inference
Upper swab wet, but unstained.	ureterovaginal fistula.
middle swab wet & stained blue.	Vesicovaginal fistula.
Only lower swab wet & stained blue.	Urethrovaginal fistula.